Reducing the raw cubic singlet
Conserved polynomials, descendants and cross spectra

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Abstract

The raw cubic derivative singlet contains several components with the same disorder. We derive exact finite-size identities that reduce its cubic energy block to a conserved polynomial and its mixed block to an energy-weighted quadratic singlet. Their linear counterterms vanish as $1/p$ in double scaling; the leading energy-block cross correlation is fixed by the known linear-residual kernel. An exhaustive endpoint certificate also removes the cubic-residual–quadratic cross spectrum. The remaining spectral problem concerns the linear residual's cross spectra with the quadratic and cubic channels.

1 The reduced physical question

Use standard CAR and the complex Gaussian number-preserving p-body Hamiltonian $H = H_s$, including overlapping index sets. Brackets mean ensemble average of normalized trace, at $\beta = \mu_{\text{chem}} = 0$. Set $L = [H, \cdot]$, $\mathsf{J}_H B = \{H, B\}/2$, and $\hbar = 1$; time is conjugate to H_s. First take $N, p \to \infty$, with

$$\lambda = p^2/N > 0, \quad q = e^{-4\lambda}, \quad \sigma^2 = \mathcal{J}_s^2/\lambda.$$

The preceding notes [1] define

$$A = \sum_i (Lc_i)^\dagger L c_i, \quad M_3 = \sum_i c_i^\dagger L^3 c_i, \quad T_3 = (M_3 + M_3^\dagger)/2,$$

$$(T, S, Z) = \mathcal{W}_3(T_3, \mathsf{J}_H A, H^3), \qquad R = T - aS - bZ,$$

where $\mathcal{W}_j$ is ordinary Gaussian *coupling* chaos, and

$$a = -\frac{(1-q)^2}{1+q^2}, \qquad b = \frac{2\lambda(1-q)(1+q^3)}{(1+q^2)(1+q+q^2)}.$$

After subtracting an explicit conserved polynomial $P_N(H, Q_f)$, our exact reduction and its double-scaled limit are

$$M_3 - P_N = R + a\mathsf{J}_H Y_N + F + \tfrac{3}{2} L Y_N - aB_S - bB_Z, \tag{1}$$

$$B_S, B_Z \longrightarrow 0 \quad \text{in ensemble Hilbert–Schmidt norm.} \tag{2}$$

Here $F = \mathcal{W}_1 T_3 - \bar{t}(Q_f)H$ and Y_N is the projected quadratic singlet. Their definitions and the proof follow. The diagonal kernels of F, Y, R have weights q, q^2, q^3 respectively; they alone do not determine the spectrum of (1).

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2 Exact first-chaos channels

Work in charge sector m, with $H = \sum_\nu H_\nu$ the orthogonal irreducible $U(N)$ decomposition. Set

$$C = w\binom{m}{p}\binom{N-m+p}{p}, \qquad C_- = C_{m-1}, \tag{3}$$

$$\kappa_\nu = w\binom{m-\nu}{p}\binom{N-m+p-\nu}{p}, \qquad \langle H_\nu^2\rangle = C\omega_\nu, \quad \sum_\nu \omega_\nu = 1. \tag{4}$$

w is the coupling covariance; it is not λ. The exact rank weights and covariance eigenvalues are the embedded-ensemble formulas [2], also used in our earlier channel calculation. Three Wick contractions give

$$\mathcal{W}_1 H^3 = \sum_\nu z_\nu H_\nu, \qquad z_\nu = 2C + \kappa_\nu. \tag{5}$$

For the Jordan block the result is

$$\mathcal{W}_1 \mathsf{J}_H A = \sum_\nu s_\nu H_\nu, \qquad s_\nu = CK_0 + \kappa_\nu J_\nu, \tag{6}$$

$$K_0 = 2p - m + mC_-/C + p^2/m, \tag{7}$$

$$J_\nu = \frac{p[p(N-m+1) - \nu(N-\nu+1)]}{(m-\nu)(N-m-\nu+1)}. \tag{8}$$

One derivation uses $\mathfrak{D}B = Q_f B - \sum_i c_i^\dagger B c_i$ and the exact identities

$$A = (2p - \mathfrak{D})H^2, \qquad T_3 = (3p - \mathfrak{D})H^3 - 3\mathsf{J}_H A.$$

The embedded rank-ν tensor obeys

$$\big(\mathfrak{D}[z_\nu(Q_f)H_\nu]\big)_m = [mz_\nu(m) - (m-p)z_\nu(m-1)]H_{\nu,m}.$$

Consequently the previously derived t_ν for $\mathcal{W}_1 T_3$ gives

$$s_\nu = \frac{(3p-m)z_\nu(m) + (m-p)z_\nu(m-1) - t_\nu(m)}{3}. \tag{9}$$

This nonsingular formula also covers the removable boundary in (6). Define deterministic sector averages and centered operators

$$\bar z = \sum_\nu \omega_\nu z_\nu, \qquad B_Z = \sum_\nu (z_\nu - \bar z)H_\nu, \tag{10}$$

$$\bar s = \sum_\nu \omega_\nu s_\nu, \qquad B_S = \sum_\nu (s_\nu - \bar s)H_\nu. \tag{11}$$

Thus $Z = H^3 - \bar z H - B_Z$. At finite size Z need not commute with H: coupling-chaos subtraction is not the same operation as forming an orthogonal polynomial in H. The latter is the q-Hermite construction in [3].

3 Counterterms and the first finite-size scale

At half filling put $s = p - \nu$. Earlier exact rank ratios give $s \Rightarrow \mathrm{Pois}(\lambda)$ with a uniform factorial tail, and $\|F\|_2^2 \to V = 64\mathcal{J}_s^6 q^2$. We now obtain

$$\boxed{\left\| pB_Z - \tfrac{1}{2}F \right\|_2 \to 0, \qquad \left\| pB_S - \tfrac{1-2\lambda}{2}F \right\|_2 \to 0.} \tag{12}$$

Here $\|B\|_2^2 = \mathbb{E}\,\mathrm{tr}(B^\dagger B)$. To see the coefficients, write $k_\nu = \kappa_\nu / C$. The exact product is

$$k_\nu = \prod_{j=0}^{p-1} \left(1 - \frac{\nu}{m-j}\right)\left(1 - \frac{\nu}{N-m+p-j}\right).$$

For each fixed s, expanding this product and the rational J gives

$$k_p \to q, \qquad p(k_{p-s} - k_p) \to 4\lambda q s, \tag{13}$$

$$J_p \to -2\lambda, \qquad p(J_{p-s} - J_p) \to 4\lambda s, \tag{14}$$

$$p\frac{s_{p-s} - s_p}{C} \to 4\lambda q(1-2\lambda)s. \tag{15}$$

The corresponding previously established slope is $(t_{p-s} - t_p)/C \to 8\lambda q s$. Center all three coefficients under the *finite* rank weights, then pass to the Poisson limit. This yields (12), including its constants.

For completeness the tail step can be controlled without a rank cutoff. In the central charge window $|m - N/2| \le N^{2/3}$, all denominators in the product and in J are bounded below by cN for sufficiently large N. For real $0 \le \nu \le p$,

$$0 \le k_\nu \le 1, \quad |\partial_\nu k_\nu| \le K_\lambda / p, \quad |J_\nu| \le K_\lambda, \quad |\partial_\nu J_\nu| \le K_\lambda / p.$$

Hence $p|k_{p-s} - k_p|$ and $p|k_{p-s}J_{p-s} - k_p J_p|$ are bounded by $K_\lambda s$. Together with the earlier bound for the t slope and $\omega_{p-s} \le K_\lambda A_\lambda^s / s!$, this proves weighted L^2 convergence and justifies centering. The expansions are uniform on this charge window. Outside it, the exact nonsingular coefficients have polynomial growth times the maximal sector variance; the binomial charge tail is exponentially small. This extends (12) to the full infinite-temperature trace.

The limit is at fixed positive λ, followed, if needed, by $\lambda \to 0^+$. No simultaneous p, λ, t scaling is asserted.

4 A nonzero cross term and its controlled limit

The conserved part of Z is $H^3 - \bar{z}(Q_f)H$. Since $\langle FZ\rangle = 0$ by ordinary Gaussian-chaos orthogonality, invariance of trace under evolution gives the exact all-time identity

$$\boxed{\langle F(t)Z\rangle = \langle FB_Z\rangle - \langle F(t)B_Z\rangle.} \tag{16}$$

It follows that $|\langle F(t)Z\rangle| \le 2\|F\|_2\|B_Z\|_2$, uniformly in real time. Thus the unscaled cross term vanishes in DS, but its leading correction is nontrivial:

$$\boxed{\lim_{\mathrm{DS}} 2p\langle F(t)Z\rangle = V[1 - C_F(t)]}, \qquad C_F(t) = \frac{\langle F(t)F\rangle_{\mathrm{DS}}}{V}. \tag{17}$$

The residual after replacing $2pB_Z$ by F in (16) is bounded by $2\|F\|_2\|2pB_Z - F\|_2$. The remaining time limit is the previously proved $r = q$ Mehler kernel. Equation (17) is therefore a fixed-time statement, obtained without differentiating a merely pointwise time limit.

The curvature follows independently. For any invariant linear Gaussian filter B, tracial Hölder and Gaussian fourth-moment bounds imply

$$\|LB\|_2^2 \le 4\|H\|_4^2\|B\|_4^2 \le 12C\|B\|_2^2.$$

Applying this to the error in (12) proves

$$2p\langle FL^2Z\rangle \longrightarrow -\langle FL^2F\rangle_{\mathrm{DS}} = -\frac{128\mathcal{J}_s^8 q^2(1-q)}{\lambda}. \tag{18}$$

No independence between B and H is used.

An exact finite example demonstrates why the cross terms had to be computed. With $N = 5, p = 2, m = 3, w = 1$, let $B_u(C_3) = \langle FH^uC_3H^{2-u}\rangle$ for $C_3 = T, S, Z$. Full polarized CAR tensors and the six allowed Wick contractions give

Block	B_0	B_1	B_2	$\langle FL^2C_3\rangle$
T	4590	5400	4590	−1620
S	−6570	−5760	−6570	−1620
Z	−4680	−3600	−4680	−2160

Here $\langle F^2\rangle = 180$ and $\langle FL^2F\rangle = 3240$. These finite-size values are not obtained by substituting $p = 2$ into a DS asymptotic.

5 The quadratic descendant and the raw operator

Define the exact sector projection

$$a_0 = \mathbb{E}A = (2p - m)C + mC_-, \qquad O = A - a_0\mathbf{1}, \tag{19}$$

$$Y_N = O - c_N(H^2 - C\mathbf{1}), \qquad c_N = \frac{\langle O(H^2 - C)\rangle}{\langle (H^2 - C)^2\rangle}. \tag{20}$$

All scalar coefficients act as functions of Q_f on full Fock space. In sectors with zero denominator set $c_N = 0$; these degenerate sectors do not contribute to the central DS limit. For nondegenerate sectors, $\langle H\mathsf{J}_H A\rangle = C\bar{s}$ and the three fourth-order Wick pairings give

$$c_N = \frac{\bar{s} - a_0}{C + \bar{\kappa}} = \frac{p^2/m + \sum_\nu \omega_\nu(\kappa_\nu/C)J_\nu}{1 + \sum_\nu \omega_\nu\kappa_\nu/C} \longrightarrow c, \tag{21}$$

$$c = \frac{2\lambda(1-q)}{1+q}, \qquad \bar{\kappa} = \bar{z} - 2C. \tag{22}$$

Because $\langle H^2 Y_N\rangle = 0$, the first chaos of its Jordan descendant has no scalar H component:

$$G_N := \mathcal{W}_1\mathsf{J}_H Y_N = B_S - c_N B_Z, \tag{23}$$

$$S = c_N Z + \mathsf{J}_H Y_N - G_N, \tag{24}$$

$$pG_N \longrightarrow \tfrac{1}{2}(1 - 2\lambda - c)F \quad \text{in } L^2. \tag{25}$$

Indeed, apply $\mathcal{W}_3$ to $\mathsf{J}_H Y_N = \mathsf{J}_H A - c_N H^3 + (c_N C - a_0)H$, and use $\bar{s} - a_0 = c_N(\bar{z} - C)$. This proves (24) exactly.

Using $T_3 = R + aS + bZ + \bar{t}H + F$ and $M_3 = T_3 + \frac{3}{2}LA$, set

$$P_N = (ac_N + b)H^3 + [\bar{t} - (ac_N + b)\bar{z}]H. \tag{26}$$

Then substitution yields (1) with no omitted finite-size terms; also $LA = LY_N$. Thus the independent energy-resolved unknowns can be organized in the basis (F, Y, R), rather than (F, S, Z, R). In DS,

$$Y = \sqrt{W_d}X, \qquad W_d = 4\mathcal{J}_s^4\left[\lambda^{-1} + \frac{4q}{1+q}\right],$$

where X is our earlier normalized quadratic singlet. In an energy basis the Y term in (1) is weighted by

$$a\frac{E + E'}{2} + \frac{3}{2}(E - E').$$

The raw M_3 is not Hermitian: the spectral Gram matrix must use the appropriate adjoint. This reduction specifies the observable before its cross spectra are combined.

6 The cubic residual decouples from the quadratic spectrum

The earlier exhaustive endpoint reduction for two cubic blocks can be reused with different weights. Put

$$D = (1+q^2)(1+q+q^2), \quad B = 2(1-q)(1+q^3), \quad (P_1, P_2, P_3) = (-q(1+q+q^2), -q(1+q+q^2), D).$$

For a pairing π, let cr count crossings, let $\alpha_i, \beta_j = 2 - 4\mathbf{1}_{\text{crossing}}$ for the marked chord pairs $12, 23, 13$ at the respective blocks, and let $\delta_{ij} = 4$ for a nested double bridge and zero otherwise. Up to common nonzero factors, the R–S orbit has coefficients

$$[\lambda]: \quad \sum_{\pi} q^{\mathrm{cr}} \sum_{i} P_i(\delta_{i1} + \delta_{i2}), \tag{27}$$

$$[\lambda^2]: \quad \sum_{\pi} q^{\mathrm{cr}} (\textstyle\sum_i P_i \alpha_i - B)(\beta_1 + \beta_2), \tag{28}$$

and R–Z has coefficient $\sum_\pi q^{\mathrm{cr}}(\sum_i P_i\alpha_i - B)$. Every one is identically zero in every endpoint template, including the fully bridged template.

To explain the all-order scope, delete spectator chords having both endpoints outside the two consecutive cubic blocks. Permuting slots within either block leaves its crossing count with every spectator unchanged. The remaining template is specified by bridge count $k = 0, 1, 2, 3$, the color order of $6 - 2k$ external endpoints, and the split between the two arcs. There are $140 + 30 + 6 + 1 = 177$ templates and 6234 pairings. Exact polynomial evaluation of these coefficients therefore proves, for every fixed $u, v \geq 0$,

$$\langle RH^u SH^v \rangle_{\mathrm{DS}} = \langle RH^u ZH^v \rangle_{\mathrm{DS}} = 0. \tag{29}$$

The local minor factors are separately checked against the existing determinant implementation in all 37404 marked cases. This shares the earlier template generator, rather than constituting a second independent enumeration proof.

There is a useful consequence for Y. For Hermitian insertions define the matrix of two-energy measures by $d\nu_{AB} = \mathbb{E}\,\mathrm{tr}[P(dE)AP(dE')B]$. It is positive as a matrix measure and satisfies spectral Cauchy–Schwarz. The diagonal R, Y limits are the absolutely continuous Mehler measures with weights q^3, q^2. Their moment bounds give tightness and uniform integrability for cross measures, supported in the same limiting compact energy square. The L^2-small linear G_N makes no fixed polynomial moment contribution (Gaussian moment bounds and Hölder). Equations (24) and (29) then imply $(E + E')d\nu_{RY} = 0$. Spectral Cauchy–Schwarz makes ν_{RY} absolutely continuous; it cannot be supported on the line $E + E' = 0$. Thus $\boxed{d\nu_{RY} = 0}$. This argument applies to every subsequential limit, establishing the full vanishing cross-spectrum limit.

7 Evidence, remaining calculation and sources

The remaining unknown energy-resolved cross measures are ν_{FR} and ν_{FY}. In particular, an ordinary time correlator that vanishes by parity would not dispose of the energy-sum-weighted F–$\mathsf{J}_H Y$ term. The next calculation should insert the centered rank filter F into the existing marked-chord contractions, keeping the first correction that survives its rank centering. The present result does not yet justify summing three diagonal probe spectra.

The new numerical records in `results/susskind-cubic-cross` contain:

- Three finite CAR systems for F against T, S, Z with two external Hamiltonians, and 32 first-chaos Jordan-channel comparisons from four (N, p) systems. The finite tensor computation uses bounded Gaussian-integer arrays and rational final normalization.
- Fifteen full-rank sums, $\lambda = 0.1, 0.25, 0.5$ and $p = 16, 32, 64, 128, 256$, repeated at 80 and 120 decimal digits. Their stored scaled-counterterm squared errors agree within 10^{-60}; this is a precision check, not interval certification.
- The exact 177-template cross certificate and 37404 local determinant-minor checks described above.
- An independent rank-bilinear calculation reproduces the three finite F–Z curvatures, including -2160. Twelve exact rational matrix identities check the Jordan, raw-operator and commutator reductions in three deterministic systems. An initial checker used structural equality on unexpanded expressions and stopped; expanding the exact residuals resolves it. The original checker is retained.

For example, at $\lambda = 0.1, p = 256$ the squared errors in the Z, S, G versions of (12), divided by $\|F\|_2^2$, are respectively 6.54×10^{-6}, 8.04×10^{-6} and 7.48×10^{-6}. At $\lambda = 0.5, p = 256$ the S error is 1.94×10^{-5}. All ranks are summed; finite-size deviations are retained, and these numbers are not errors on a bulk prediction.

The known covariance and q-Hermite constructions below are used as inputs, not claimed as new results. The new work in this note is the explicit reduction for these same-coupling derivative insertions and its stated cross-spectrum consequences. It remains a model calculation within the Susskind singlet program: bulk locality, masses and the finite-gravity interaction dictionary require further observables.

References

[1] A. Day and Chengheng, local research notes (2026): *An independent cubic derivative singlet*; *The missing linear component of the cubic singlet*; *Dynamics of the linear cubic residual*; *The linear cubic residual has a full q kernel*; and the quadratic/cubic full-kernel notes. Source directories: `manuscript/susskind-cubic-*`, `susskind-linear-kernel`, `susskind-singlet-kernel`.

[2] V. K. B. Kota, *Bivariate moments of the two-point correlation function for embedded Gaussian unitary ensemble with k-body interactions*, arXiv:2208.11312v2, Sec. IV, Eqs. (28)–(31).

[3] J. Tall and S. Tomsovic, *Double-scaled bosonic and fermionic embedded ensembles, complex SYK, and the dual Hilbert space*, arXiv:2604.14522v2, Sec. 3.1, Eqs. (3.1)–(3.14), and Sec. 4.1.